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Patient-Reported Outcomes

A Quantitative Exploration of the Performance of a Potential “Health and Well-Being” Visual Analog Scale (EQ-HWB VAS) for the EQ-HWB-9

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ABSTRACT

Objectives: All EuroQol instruments include a visual analog scale (VAS), except the EuroQol Health and Well-Being (EQ-HWB), which captures broader aspects of well-being beyond health, relevant to populations providing and receiving care. This study explored the performance of a potential VAS for EQ-HWB.

Methods: The EQ-HWB VAS was developed by adapting the EQ VAS from the EuroQol 5-Dimension 5-Level instrument (EQ-5D-5L), replacing “health” with “health and well-being” and “today” with “last week” to align the recall period with EQ-HWB. A cross-sectional survey was undertaken with an age- and sex-representative sample of the Hong Kong general population ($n = 1262$). Descriptive characteristics, ceiling and floor, convergent and known-group validity, and explanatory power were assessed. The performance of the EQ-HWB VAS was directly compared with the EQ VAS.

Results: The mean EQ VAS and EQ-HWB VAS scores were 76.7 and 73.5 ($P < .001$). The ceiling was 5.2% for the EQ VAS and 3.6% for EQ-HWB VAS. The EQ-HWB VAS showed a very strong correlation and agreement with EQ VAS. Both the EQ VAS and EQ-HWB VAS were able to distinguish between most known groups. The EQ-HWB VAS showed larger effect sizes for groups such as caregiving, perceived income level and mental health problems. EQ-HWB-9 items explained more variance in EQ-HWB VAS scores than EQ-5D-5L dimensions did in EQ VAS scores.

Conclusions: This study provides the first evidence of the measurement properties of a “health and wellbeing” VAS for EQ-HWB, demonstrating its validity. Further research is recommended in different languages and countries to inform decisions about the addition of a VAS to the EQ-HWB.

Keywords: EQ-5D-5L, EQ-HWB-9, health, measurement properties, VAS, well-being.

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Highlights

- The EuroQol visual analog scale (EQ VAS) is a useful component of the EuroQol 5-Dimension instrument, demonstrating good psychometric performance across various conditions and cultures. The EuroQol Health and Well-Being (EQ-HWB), an experimental EuroQol measure, extends beyond health and includes aspects of well-being; however, it lacks a VAS.
- This study introduces and validates a novel EQ-HWB VAS adapted from the EQ VAS with a “health and well-being” label. We demonstrate its robust validity, including high agreement with EQ VAS, good convergent and known-group validity, providing the first empirical evidence of its measurement properties.
- The EQ-HWB VAS could enhance the EQ-HWB and EQ-HWB-9 by providing a single numeric value that captures broader well-being aspects in addition to health. Further qualitative and quantitative validation is required across diverse populations, countries, and languages.

Introduction

Originating in psychophysics, the visual analog scale (VAS) was initially used to quantify responses to sensory stimuli, such as light, sound, or heat.¹ Over time, the use of VAS diffused into social and health sciences, as well as medicine to assess subjective experiences, such as emotions, attitudes, pain, mood, or satisfaction, by asking respondents mark a point on a continuous line, with anchors at each end (eg, “no pain” to “worst possible pain”). The distance from one end is measured to quantify the response, often as a score from 0 to 10 or 0 to 100, the latter potentially offering higher sensitivity to smaller changes.² VAS is widely used in clinical and research settings, particularly for pain assessment in conditions such as chronic pain, cancer, or post-operative recovery because of its simplicity, validity, and reliability across diverse populations.³⁻⁷ Furthermore, in the 1970s, VAS evolved into a tool for eliciting preferences for health states.^{8,9} The EQ VAS was originally part of the EuroQol 5-Dimension instrument (EQ-5D) as a warm-up for VAS valuation tasks. It later gained recognition as a self-reported global

measure of overall health, complementing the descriptive system, and became a standard component of the instrument.¹⁰

Currently, all EuroQol instruments, whether approved or experimental, adult (EQ-5D) and youth (EQ-5D-Y and EQ-TIPS), include an EQ VAS that provides valuable numeric data in patient and population surveys, clinical trials, and psychometric evaluations of the EQ-5D descriptive system.¹¹⁻¹³ This is a vertical, hash-marked health VAS with a “today” recall period, designed to align with the descriptive system. It uses endpoints of “the best health you can imagine” (100) and “the worst health you can imagine” (0).¹⁴ The EQ VAS has been reported to demonstrate good psychometric performance across various health conditions and cultural contexts.¹⁵ However, no VAS has been developed for the EQ Health and Well-being (EQ-HWB) yet. The EQ-HWB is currently an experimental EuroQol measure, designed to capture quality of life related to health, social care and caregiving, thus

encompassing broader aspects than health alone.¹⁶ It has 2 versions: a long version (EQ-HWB), which functions as a profile measure, and a shorter, 9 item, EQ-HWB-9, which incorporates a subset of items and is intended as a generic preference-accompanied measure.

Although the EQ VAS offers a valid and reliable self-reported measure of overall health, it is limited to a health-focused frame and may not fully capture broader well-being aspects (eg, loneliness and control), covered by the EQ-HWB instrument. Adapting the VAS to reference “health and well-being” explicitly could provide a complementary summary score aligned with EQ-HWB’s scope, enabling broader outcome evaluations in contexts in which the measurement of well-being in addition to health is relevant. Furthermore, the EQ VAS has a “today” recall period that does not align with the EQ-HWB’s 7-day recall. Similar to the value of the EQ VAS in EQ-5D instruments, a 7-day “health and well-being” VAS may provide additional value to users across various sectors (eg, caregivers, social care) for which the instrument is intended to be used. Therefore, this study aimed to explore the measurement performance of a potential EQ-HWB VAS and compare it with the EQ VAS from the EQ-5D-5L.

Methods

Study Design and Participants

This study used data from a web-based cross-sectional survey, conducted among members of Hong Kong general population between January and March 2025. Participants were recruited through the online panel of a local surveying company. Inclusion criteria were as follows: (1) aged between 20 and 69, (2) able to read and speak Chinese, and (3) able to provide informed consent. The exclusion criteria were as follows: (1) failure to complete the CAPTCHA questions, (2) identification as bots by the Qualtrics platform, (3) completion of the questionnaire in less than 6 minutes or more than 30 minutes, (4) duplicate IP addresses, and (5) failure to provide consistent responses to cross-validated questions. We aimed to obtain a representative sample using quota sampling based on sex and age. To ensure the integrity of the data set, all questions were made mandatory, preventing respondents from proceeding to the next question without answering the previous one. During data collection, quality control measures were applied in real time. This approach ensured that exclusions refined the sample while preserving its overall representativeness because the initial recruitment frame was designed to be demographically balanced from the outset. The study protocol was approved by the ethics committee of the Institutional Review Board of the *blind for review* (Ref No.: HSEARS20240917004), and written informed consent was collected from all participants.

The questionnaire was programmed in Qualtrics (Provo, UT, USA, 2020). The questionnaire, including all outcome measures, was administered in Traditional Chinese. The following outcome measures were included in the survey: EQ-5D-5L, EQ-HWB-9, PROMIS-29+2, and ReQoL-10. The latter 2 were selected for allowing convergent validity testing. The rationale for this selection was to include measures that capture both health-related and broader aspects, similar to the EQ-HWB-9, eg, social domains in PROMIS-29+2, and loneliness and control in ReQoL-10.

Outcome Measures Included in the Survey

EuroQol instruments

All EuroQol instruments and their modifications were registered, and permission to modify agreement was signed with the

EuroQol Office. This study included the EQ-5D-5L¹⁷ and the experimental version of EQ-HWB-9¹⁶ instruments, completed in a randomized order. An EQ-HWB VAS was developed, mirroring the EQ VAS layout and format, but with revised wording to reflect “health and well-being” (Traditional Chinese: 身心健康) instead of only “health.” The scale ranged from the best to the worst health and well-being you can imagine, and the recall period was changed from “today” to “the last 7 days” to align with the EQ-HWB-9 descriptive system. We have chosen “health and well-being” VAS because it is in the name of the instrument and already appears in the instruction line, therefore a validated and cognitively debriefed translation of the term was available in Traditional Chinese. The EQ-HWB VAS was incorporated after the EQ-HWB-9 descriptive system. For both EuroQol instruments, we calculated level sum scores (LSSs) transformed to a 0 to 100 scale (higher scores indicating better outcomes) and utility values. The EQ-5D-5L utility values were estimated using the HK (−0.863 to 1) and UK (Hernández et al) crosswalk (−0.594 to 0.99) value sets.^{18,19} For the EQ-HWB-9, utility values were computed using the UK pilot value set, which was estimated based on the same EQ-HWB-9 version (experimental, 2022).²⁰

Patient-Reported Outcomes Measurement Information System (PROMIS)-29+2

PROMIS-29+2 measures health with 31 items across 8 domains.²¹ A 7-day recall period is used for most of its domains, except for physical function and ability to participate in social roles and activities, which do not specify a recall period. We computed PROMIS-Preference (PROPr) score utilities ranging from −0.022 to 0.954, using the US value set.²²

Recovering Quality of Life (ReQoL)-10

The ReQoL-10, a 10-item instrument, designed to assess quality of life in individuals with mental health conditions using a 1-week recall period, with higher scores reflecting better quality of life.²³ The ReQoL-Utility Index (ReQoL-UI) uses 7 items to generate a utility value, using the HK value set (−0.814 to 1) in this study.²⁴ LSSs were also computed.

Background information

Participants provided information on chronic disease status (“In the past 12 months, have you had any chronic medical conditions or been diagnosed with a medical condition by a doctor?”). If respondents answered “Yes,” they were directed to a multiple-choice question listing over 10 prevalent chronic diseases, with a text box to specify any conditions not included in the list. Further questions included general health status (5-point scale from “very poor” to “excellent”), demographics (eg, sex), socioeconomic factors (eg, income, caregiver status), and lifestyle behaviors (eg, healthy diet).

Structure of the Survey

The survey began with an informed consent form on the first page, which participants were required to read and agree to before proceeding. Four variants of EQ instruments, EQ-5D-5L descriptive system with EQ VAS (“today” and “last 7 days” time frames) and EQ-HWB-9 descriptive system with EQ-HWB VAS (“today” and “last 7 days” time frames), were incorporated in the survey. The questionnaire was structured into 4 sections. Section 1 presented 2 of the 4 variants in randomized order: either an EQ-5D-5L (any time frame) followed by an EQ-HWB-9 (any time frame), or vice versa. Section 2 included PROMIS-29+2 and ReQoL-10 in a randomized sequence. Section 3 presented the 2 remaining variants of EQ-5D-5L and EQ-HWB-9 descriptive

systems with their respective VAS versions, also in a randomized order. Section 4 collected background information. Analyses of the impact of recall period on the descriptive system are reported elsewhere.²⁵ This manuscript focuses solely on the EQ VAS ("today") and EQ-HWB VAS ("last week"), which followed the EQ-5D-5L descriptive system ("today") and EQ-HWB-9 descriptive system ("last 7 days"), respectively, because these represent the standard recall period versions of the 2 instruments. To aid interpretation of the results, we report descriptive characteristics for the other 2 VAS versions as well.

Data Analysis

Descriptive characteristics

The representativeness of the sample based on sex and age was checked against the HK general population for key socio-demographic variables using Chi-squared test. The relative frequency of responses for each response level of each dimension/item in the descriptive systems of the EQ-5D-5L and EQ-HWB-9 was computed. Descriptive statistics for the EQ VAS and EQ-HWB VAS scores were computed, including the mean (standard deviation), median (interquartile range), and range of scores. Histograms were used to visualize the distribution of EQ VAS and EQ-HWB VAS scores, as well as their mean differences. Ceiling and floor were defined as the proportion of respondents reporting the best and worst VAS scores, respectively. Mean differences in scores between EQ VAS and EQ-HWB VAS were computed and compared using paired *t* test. We hypothesized that, on average, EQ-HWB VAS scores would be lower than EQ VAS scores, with a correspondingly lower ceiling for the EQ-HWB VAS than for the EQ VAS. This is due to the broader aspects captured by the EQ-HWB VAS anchors, the longer recall period and the potential priming effect of the preceding 9 items of EQ-HWB-9 versus the 5 dimensions of EQ-5D-5L.

Convergent validity and agreement

Convergent validity was assessed by testing hypotheses about correlations between EQ-5D-5L LSSs/utility, EQ-HWB-9 LSSs/utility, EQ VAS, and EQ-HWB VAS with each other, as well as with PROPr and ReQoL-UI. We used Pearson's correlation coefficients, categorized as weak (<0.3), moderate (0.3 – 0.5), or strong (>0.5) to assess convergent validity.²⁶ The agreement between the EQ VAS and EQ-HWB VAS was assessed using an intraclass correlation coefficient (ICC; 2-way mixed-effects model, absolute agreement, >0.7 good reliability)²⁷ and Bland-Altman plots.²⁸ We expected strong correlations and agreement between EQ VAS and EQ-HWB VAS, as well as at least moderate correlations with the LSSs or utilities of EQ-5D-5L, EQ-HWB-9, PROPr, and ReQoL-UI.^{29–31}

Known-group validity

Independent samples *t* tests (2 groups) and 1-way analysis of variance (more than 2 groups) were used to evaluate whether EQ VAS and EQ-HWB VAS were able to distinguish between pre-defined known groups of respondents. Cohen's *d* (2-group comparison, ≥ 0.8 large, 0.5 – 0.79 medium, 0.2 – 0.49 , small, and <0.2 , trivial) and η^2 (3-group comparison, ≥ 0.14 large, 0.06 – 0.13 medium, and 0.01 – 0.05 , small) were calculated to reflect the effect size (ES).³² We hypothesized that individuals with worse general health status, chronic conditions, informal caregiving responsibilities, lower income, or unhealthy lifestyle behaviors (eg, poor diet, or insufficient exercise) would report lower health and well-being. The analysis also included 3 PROMIS-29+2 items, each representing the first item in its respective domain: sleep quality (Sleep 109, "my sleep quality was..."), fatigue (HI7, "I feel

fatigued"), and pain interference (PAININ9, "how much did pain interfere with your day-to-day activities?"). For chronic conditions, each disease group was compared with healthy respondents.

Regression analyses on EQ VAS and EQ-HWB VAS

Multivariable linear regression models were used to examine how EQ-5D-5L, EQ-HWB-9, and PROMIS-29+2 explain variance in EQ VAS, EQ-HWB VAS, and difference between the 2 VAS scores (EQ VAS vs EQ-HWB VAS). Six models were developed, using the 5 dimensions from EQ-5D-5L, the 9 items from EQ-HWB-9, and the 8 domains from PROMIS-29+2, respectively, adjusting for covariates including age, sex, education, income, smoking, alcohol consumption, weekly exercise duration, diet, sleep quality, and presence of chronic disease. Adjusted R-squared was used to assess explanatory power. A bootstrapping approach with 1000 resamples to compute 95% confidence intervals was used to evaluate the variability and robustness of the adjusted R-squared values. Additionally, we explored the determinants of the difference between EQ VAS and EQ-HWB VAS scores. The dependent variable was this difference, and all socio-demographic and health-related variables were included as independent variables. The Shorrocks-Shapley decomposition analysis was used to evaluate the relative contribution of the EQ-5D-5L dimensions, EQ-HWB-9 items, and PROMIS-29+2 domains in explaining the variance of EQ VAS and EQ-HWB VAS.³³

Results

Participants' Background Characteristics

In total, 1262 participants completed the survey and provided valid responses. Of the 2168 who clicked the survey link, 1682 finished it, but 420 were excluded for failing to meet data quality criteria. Females comprised 55.1% of the final sample, approximately 40% were over 50 years old, and 25.7% reported a diagnosed chronic condition. The sample's sex, age, and marital status distributions were comparable to the Hong Kong general population; however, highly educated respondents were over-represented (Table 1).

Descriptive Characteristics

Table 2 presents the response distributions for the EQ-5D-5L and EQ-HWB-9. For the EQ-5D-5L, the proportion of respondents reporting the best possible response (ceiling) ranged from 49.0% (pain/discomfort) to 91.4% (self-care). For the EQ-HWB-9, the ceiling ranged from 32.1% (exhausted) to 83.7% (getting around inside and outside).

The mean EQ VAS score (76.72) was significantly higher than the mean EQ-HWB VAS score (73.49) ($P < .001$). The ceiling on the EQ VAS was slightly higher (5.2%) than on the EQ-HWB VAS (3.6%). Overall, 18.4% showed identical, 55.5% lower, and 26.1% higher EQ-HWB VAS scores compared with their EQ VAS scores. The distributions of EQ VAS and EQ-HWB VAS responses were very similar, with typical clusters observed at scores of 70, 80, and 90 (Appendix Fig. 1). Descriptive statistics for the other 2 VAS versions (EQ-5D-5L "last week" and EQ-HWB-9 "today") are reported in Appendix Table 1 and Appendix Figures 2 and 3.

Convergent Validity and Agreement

The convergent validity of the EQ VAS and EQ-HWB VAS by examining their correlations with various measures is presented in Table 3. The association between EQ VAS and EQ-HWB VAS was strong with a correlation coefficient of 0.88. The EQ-HWB

Table 1. Sample characteristics.

Characteristics	N (%)	HK general population (%)	Difference	Chi-squared test
Age groups (years)				<i>P</i> = .89
20-24	78 (6.2)	6.1	0.1	
25-29	114 (9.0)	8.5	0.5	
30-34	129 (10.2)	9.8	0.4	
35-39	155 (12.3)	10.9	1.4	
40-44	143 (11.3)	10.9	0.4	
45-49	127 (10.1)	10.8	−0.7	
50-54	130 (10.3)	10.7	−0.4	
55-59	138 (11.0)	11.7	−0.8	
60-64	135 (10.7)	11.4	−0.7	
65-69	113 (9.0)	9.2	−0.2	
Gender				<i>P</i> = .92
Male	567 (44.9)	45.6	−0.7	
Female	695 (55.1)	54.4	0.7	
Marital status				<i>P</i> = .22
Married	740 (58.6)	58.6	0	
Never married	458 (36.3)	28.6	7.7	
Divorced/Widowed	64 (5.1)	12.7	−7.6	
Educational level				<i>P</i> < .001
Primary and below	21 (1.7)	18.4	−16.7	
Secondary	493 (39.1)	47	−7.9	
Post-secondary	748 (59.3)	34.6	24.7	
Perceived income level				
Below local average	234 (18.5)			
Local average	764 (60.5)			
Above local average	264 (20.9)			
Employment				
Full-time	933 (73.9)			
Part-time	113 (9.0)			
Others	216 (17.1)			
Living condition				
Alone	146 (11.6)			
With others	1116 (88.4)			
Informal caregiver				
Yes	323 (25.6)			
No	939 (74.4)			
Chronic diseases				
Yes	324 (25.7)			
No	829 (65.7)			
Don't want to answer the question	109 (8.6)			

Note. HK general population data: <https://www.census2021.gov.hk/en/index.html>
 Accessed: January 2, 2024

VAS demonstrated moderate correlations with the EQ-5D-5L LSS and HK utility (both 0.43), and strong correlations with EQ-5D-5L UK utility (0.52), EQ-HWB-9 LSS (0.63), EQ-HWB-9 utility (0.60), ReQoL-10 LSS (0.58), ReQoL utility (0.56), and PROPr utility (0.49). Comparatively, the EQ VAS exhibited higher correlations with the EQ-5D-5L LSS and HK utility (both 0.51) and with the EQ-5D-5L UK utility (0.70), while showing similar correlations to the EQ-HWB VAS for EQ-HWB-9 LSS, EQ-HWB-9 utility, ReQoL-10 LSS and utility, and PROPr utility.

The Bland-Altman plot and ICC value (Fig. 1) demonstrated good agreement between the EQ VAS and EQ-HWB VAS, with an ICC of 0.88. At both the high and low ends of the scale, EQ VAS and EQ-HWB VAS scores were more closely aligned. In contrast, a greater dispersion of points was observed between scores of 40 and 80, indicating increased variability. This may suggest the presence of subgroups of respondents who report good health on the EQ VAS but lower health and well-being on the EQ-HWB VAS,

or vice versa, good health and well-being on the EQ-HWB VAS but poorer health on the EQ VAS.

Known-Group Validity

All hypotheses were supported, with both EQ VAS and EQ-HWB VAS scores being lower in the groups as expected (Table 4). EQ VAS scores were higher in all subgroups compared with EQ-HWB VAS scores, with the largest difference observed in those with anxiety/depression (EQ VAS: 55.45, EQ-HWB VAS: 50.38). Both the EQ VAS and EQ-HWB VAS were able to significantly distinguish between all known groups. Among the known groups in which a significant difference was observed, EQ VAS had higher effect sizes for 7 groups, whereas EQ-HWB VAS had higher effect sizes for 6 groups. The ES was identical for 1 group. The EQ VAS showed a slightly larger ES for typical health-related variables, such as general health status, chronic diseases, hypertension, hyperlipidemia, visual problems,

Table 2. Descriptive characteristics of the EQ-5D-5L and EQ-HWB-9.

Instruments	Levels [N (%)]				
	1	2	3	4	5
EQ-5D-5L					
Mobility	1079 (85.5)	141 (11.2)	36 (2.9)	6 (0.5)	0 (0.0)
Self-care	1154 (91.4)	79 (6.3)	25 (2.0)	4 (0.3)	0 (0.0)
Usual activities	1094 (86.7)	125 (9.9)	31 (2.5)	9 (0.7)	3 (0.2)
Pain/Discomfort	618 (49.0)	545 (43.2)	83 (6.6)	14 (1.1)	2 (0.2)
Anxiety/Depression	673 (53.3)	438 (34.7)	103 (8.2)	30 (2.4)	18 (1.4)
11111	439 (34.8)	-	-	-	-
EQ-HWB-9					
Getting around inside and outside	1056 (83.7)	154 (12.2)	38 (3.0)	11 (0.9)	3 (0.2)
Day-to-day activities	1011 (80.1)	192 (15.2)	46 (3.6)	12 (1.0)	1 (0.1)
Exhausted	405 (32.1)	563 (44.6)	208 (16.5)	75 (5.9)	11 (0.9)
Lonely	518 (41.0)	476 (37.7)	194 (15.4)	55 (4.4)	19 (1.5)
Concentrating/thinking clearly	515 (40.8)	493 (39.1)	198 (15.7)	47 (3.7)	9 (0.7)
Anxious	488 (38.7)	459 (36.4)	208 (16.5)	82 (6.5)	25 (2.0)
Sad/Depressed	563 (44.6)	420 (33.3)	203 (16.1)	49 (3.9)	27 (2.1)
Control	578 (45.8)	419 (33.2)	178 (14.1)	67 (5.3)	20 (1.6)
Pain (severity)	481 (38.1)	656 (52.0)	112 (8.9)	10 (0.8)	3 (0.2)
111111111	184 (14.6)	-	-	-	-
Visual analog scales	Mean (SD)	Median (IQR)	Range	Ceiling	Difference of mean
EQ VAS	76.72 (17.68)	80.00 (21.00)	(1.00, 100.00)	5.2%	3.23***
EQ-HWB VAS	73.49 (18.38)	77.00 (24.00)	(0.00, 100.00)	3.6%	
EQ VAS = EQ-HWB VAS (N%)	232 (18.4)				
EQ VAS > EQ-HWB VAS (N%)	700 (55.5)				
EQ VAS < EQ-HWB VAS (N%)	330 (26.1)				

Note. The range of the EQ VAS and EQ-HWB VAS indicates the lowest scores reported by participants in this study. The theoretical range for both scales is 0 to 100. *** $p < .001$.

and skin diseases. The most notable difference was observed for skin diseases (EQ VAS: $d = 0.94$ vs EQ-HWB VAS: $d = 0.85$). Conversely, EQ-HWB VAS exhibited slightly larger ES in groups associated with well-being, including caregiver status, perceived income level, exercise status, and mental health problems.

Explanatory Power Analysis

Adjusted R^2 values were (Appendix Table 2): EQ VAS, 0.459 (EQ-5D-5L dimensions) vs 0.454 (EQ-HWB-9 items); EQ-HWB VAS, 0.427 (EQ-5D-5L dimensions) vs 0.474 (EQ-HWB-9 items). Models with PROMIS-29+2 domains as independent variables yielded similar adjusted R^2 values (EQ VAS: 0.462; EQ-HWB VAS: 0.471). Despite differences in item content, PROMIS-29+2 and EQ-HWB-9 instruments explained similar variance in EQ-HWB

VAS, suggesting that the EQ-HWB VAS captures broader aspects of health and well-being beyond the constructs measured by the EQ-HWB-9 and PROMIS-29+2.

The Shorrocks-Shapley decomposition analysis showed that the anxiety/depression was the most important EQ-5D-5L dimensions in this sample, followed by pain/discomfort (Fig. 2). EQ-HWB-9 item, exhausted, significantly affects 2 VAS scores (approximately 35% for EQ VAS and 30% for EQ-HWB VAS). EQ-HWB-9 pain (severity) was a key factor in both models and sadness/depression influenced EQ-HWB VAS more than EQ VAS. The EQ-5D-5L item anxiety/depression affected both VAS scores (EQ VAS: $\sim 70\%$; EQ-HWB VAS: $\sim 75\%$), but EQ-HWB-9 anxiety had minimal impact (less than 3%). Nonhealth items, such as control and loneliness, affected both VAS scores, with control affecting EQ VAS ($\sim 20\%$) more and lonely affecting EQ-HWB VAS ($\sim 20\%$) more. In the PROMIS models, sleep and fatigue (around 25% each) demonstrated a greater contribution than the other domains. Physical function affected the EQ VAS more than the EQ-HWB VAS, whereas social roles and activities showed the reverse pattern.

In the multivariable regression of the difference between EQ VAS and EQ-HWB VAS scores (Appendix Table 3), age showed a significant negative association (-0.08 ; 95% CI: -0.12 to -0.04), indicating that the gap between these 2 measures narrows as individuals grow older. Higher income levels (local average: -1.62 , 95% CI: -3.09 to -0.15 ; above average: -1.92 , 95% CI: -3.73 to -0.11) were also significantly associated with reduced differences, suggesting that individuals with higher incomes experience a smaller gap between EQ VAS and EQ-HWB VAS scores.

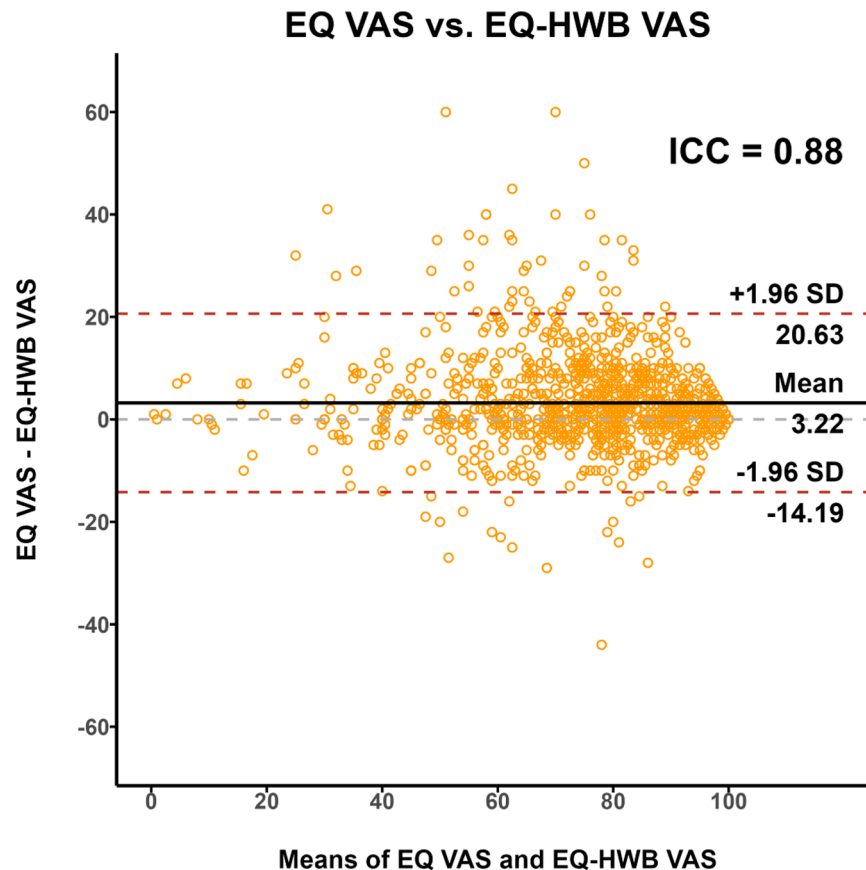
Discussion

This study examined the measurement properties of a newly developed “health and well-being” VAS for the EQ-HWB-9 and

Table 3. Convergent validity between VAS scores, utility values, and level summary scores.

Measurement	EQ-HWB VAS	EQ VAS
EQ VAS	0.88	-
EQ-5D-5L LSS	0.43	0.51
EQ-5D-5L HK utility	0.43	0.51
EQ-5D-5L UK utility	0.52	0.70
EQ-HWB-9 LSS	0.63	0.62
EQ-HWB-9 utility	0.60	0.60
ReQoL-10 LSS	0.58	0.56
ReQoL utility	0.56	0.57
PROPr utility	0.49	0.50

Note. The P value for all correlation coefficients is $<.001$. Correlation strength: >0.5 Strong; 0.31 - 0.5 Moderate; <0.3 Weak.

Figure 1. Bland-Altman plot of EQ VAS and EQ-HWB VAS scores.

compared its performance with the EQ VAS in a general population sample in Hong Kong. Including a VAS in the EQ-HWB instrument could enhance alignment with other EuroQol instruments, providing a simple metric to capture holistic self-assessments of health and well-being, extending beyond traditional health to include social care and caregiving impacts. To our knowledge, this is the first study to quantitatively test a potential VAS for EQ-HWB. Overall, the EQ-HWB VAS showed generally strong measurement properties and slightly lower ceiling and similar convergent and known-group validity compared with the EQ VAS.

We found that over half the sample had a lower EQ-HWB VAS score than their EQ VAS score, and the mean score of the EQ-HWB VAS was significantly lower than that of the EQ VAS, with the difference between them decreasing with age. However, in the total sample, the average difference was just slightly above 3 points, which is not substantial. The, on average, lower EQ-HWB VAS score may stem from a combination of various factors, including the following: (1) the broader scope of the label “health and well-being” compared with “health” alone, (2) the longer recall period (7 days vs today), (3) the differences in length and content of the questions asked before the VAS, which might prime respondents to consider broader aspects when completing the VAS. Results from the other EQ VAS (“last 7 days”) and EQ-HWB (“today”) versions, although not the focus of this article, support this interpretation. Mean EQ VAS scores were 76.72 (“today”) and 75.30 (“last 7 days”), and mean EQ-HWB VAS scores were 75.11 (“today”) and 73.49 (“last 7 days”) (Appendix Table 1). Lower scores with the “last 7 days” recall and consistently lower EQ-HWB VAS scores compared

with EQ VAS suggest that both recall period and scale anchor contribute to the observed differences.

It is well documented that interpretations of the EQ VAS endpoints (best and worst health you can imagine) vary across individuals and cultures. Individuals may not always strictly adhere to the 5 health areas of the EQ-5D (eg, they may consider cognitive functioning, sleep, and vitality) or focus not solely on health aspects but may instead include broader constructs, such as, social relationships, financial well-being, happiness, satisfaction with life, or harmony.³⁴⁻³⁶ Our results also align with this because the Shorrock-Shapley decomposition of EQ-HWB-9 items, including broader aspects than health, contributed substantially not only to EQ-HWB VAS but also to the EQ VAS scores. We need to notice that the contributions of mental and social items in the EQ-HWB-9 to both VAS outcomes may reflect a redistribution of variance captured by the broader anxiety/depression dimension in the EQ-5D-5L models.

Additionally, although the differences in adjusted R^2 values across models were small, they may indicate that EQ-HWB-9 may capture more variance in EQ-HWB VAS compared with EQ-5D, whereas PROMIS-29+2 performed comparably to EQ-HWB-9 for EQ-HWB VAS and slightly better for EQ VAS relative to the other instruments. However, we need to interpret differences in Shapley value contributions across these models with caution because the contributions represent proportions of the explained variance, which varies slightly between models. Thus, apparent differences in importance may partly reflect these underlying differences in total explained variance rather than absolute predictor strength alone.

Table 4. Known-group validity of EQ VAS and EQ-HWB VAS.

Known groups	N	EQ VAS	EQ-HWB VAS
General health status			
Excellent	112	91.66 (9.64)	88.51 (14.05)
Very good	454	83.66 (12.40)	80.79 (13.49)
Good	382	76.46 (14.40)	72.92 (15.11)
Fair	278	64.44 (17.17)	60.87 (17.05)
Poor	36	40.14 (21.73)	38.28 (21.04)
F-statistics		167.40	151.50
P value		<.001	<.001
η^2		0.35	0.33
Informal caregiver			
Yes	323	73.84 (19.01)	69.92 (19.44)
No	939	77.71 (17.10)	74.72 (17.84)
t statistics		-3.41	-4.08
p-value		<.001	<.001
Cohen's d		0.22	0.26
Perceived income level			
Below local average	234	66.83 (21.25)	62.21 (21.12)
Local average	764	77.73 (16.06)	74.70 (16.64)
Above local average	264	82.55 (15.04)	80.02 (16.14)
F statistics		56.84	69.17
P value		<.001	<.001
η^2		0.08	0.10
PROMIS-29+2 My sleepquality was			
Very poor	116	64.75 (23.19)	60.06 (19.67)
Poor	403	65.30 (20.35)	62.87 (23.54)
Fair	507	74.14 (16.00)	71.07 (16.26)
Good	190	83.66 (13.33)	80.76 (14.67)
Very good	46	87.97 (13.20)	85.09 (15.68)
F-statistics		239.1	256.8
P value		<.001	<.001
η^2		0.16	0.17
PROMIS-29+2 How much did pain interfere with your day-to-day activities			
Not at all	590	83.13 (14.74)	79.63 (16.37)
A little bit	494	74.12 (16.55)	70.58 (17.37)
Somewhat	137	63.60 (17.53)	62.43 (16.88)
Quite a bit/Very much	41	59.44 (25.47)	57.27 (25.83)
F-statistics		238.3	206
P value		<.001	<.001
η^2		0.16	0.14
PROMIS-29+2 "I feel fatigued"			
Not at all	297	87.00 (12.06)	84.95 (13.37)
A little bit	606	78.91 (13.80)	75.61 (14.45)
Somewhat	235	68.07 (18.66)	63.14 (18.52)
Quite a bit	101	60.09 (20.38)	57.36 (20.62)
Very much	23	47.30 (17.93)	46.53 (25.14)
F-statistics		422.5	425.3
P value		<.001	<.001
η^2		0.25	0.26
Diet			
Very healthy	89	88.87 (13.51)	86.64 (16.68)
Healthy	334	83.54 (13.93)	80.47 (15.25)
Fairly healthy	442	77.67 (15.13)	74.54 (15.44)
Average	366	68.49 (18.36)	64.79 (18.44)
Unhealthy	31	51.84 (23.44)	48.45 (22.73)
F-statistics		71.47	71.90
P value		<.001	<.001
η^2		0.19	0.19
Exercise			
≤90 minutes/week	498	72.93 (19.77)	69.29 (19.74)
>90 minutes/week	764	79.18 (15.70)	76.23 (16.89)
t statistics		-6.23	-6.67
P value		<.001	<.001

continued on next page

Table 4. Continued

Known groups	N	EQ VAS	EQ-HWB VAS
Cohen's d		0.36	0.38
Living with chronic conditions			
Yes	324	71.25 (20.15)	67.80 (20.41)
Healthy	829	79.57 (16.08)	76.35 (17.22)
t statistics		-7.33	-7.19
P value		<.001	<.001
Cohen's d		0.48	0.47
Hypertension			
Yes	78	71.59 (20.64)	68.87 (21.43)
Healthy	829	79.57 (16.08)	76.35 (17.22)
t statistics		-4.08	-3.59
P value		<.001	<.001
Cohen's d		0.48	0.42
Hyperlipidemia			
Yes	65	69.72 (20.75)	66.31 (20.89)
Healthy	829	79.57 (16.08)	76.35 (17.22)
t statistics		-4.64	-4.45
P value		<.001	<.001
Cohen's d		0.60	0.57
Visual problems			
Yes	40	68.40 (22.38)	65.60 (21.57)
Healthy	829	79.57 (16.08)	76.35 (17.22)
t statistics		-4.20	-3.81
P value		<.001	<.001
Cohen's d		0.68	0.62
Skin diseases (eg, psoriasis, eczema, urticaria)			
Yes	35	64.14 (24.40)	61.34 (26.71)
Healthy	829	79.57 (16.08)	76.35 (17.22)
t statistics		-5.42	-4.92
P value		<.001	<.001
Cohen's d		0.94	0.85
Anxiety/depression			
Yes	40	55.45 (23.51)	50.38 (22.57)
Healthy	829	79.57 (16.08)	76.35 (17.22)
t statistics		-9.04	-9.17
P value		<.001	<.001
Cohen's d		1.46	1.49

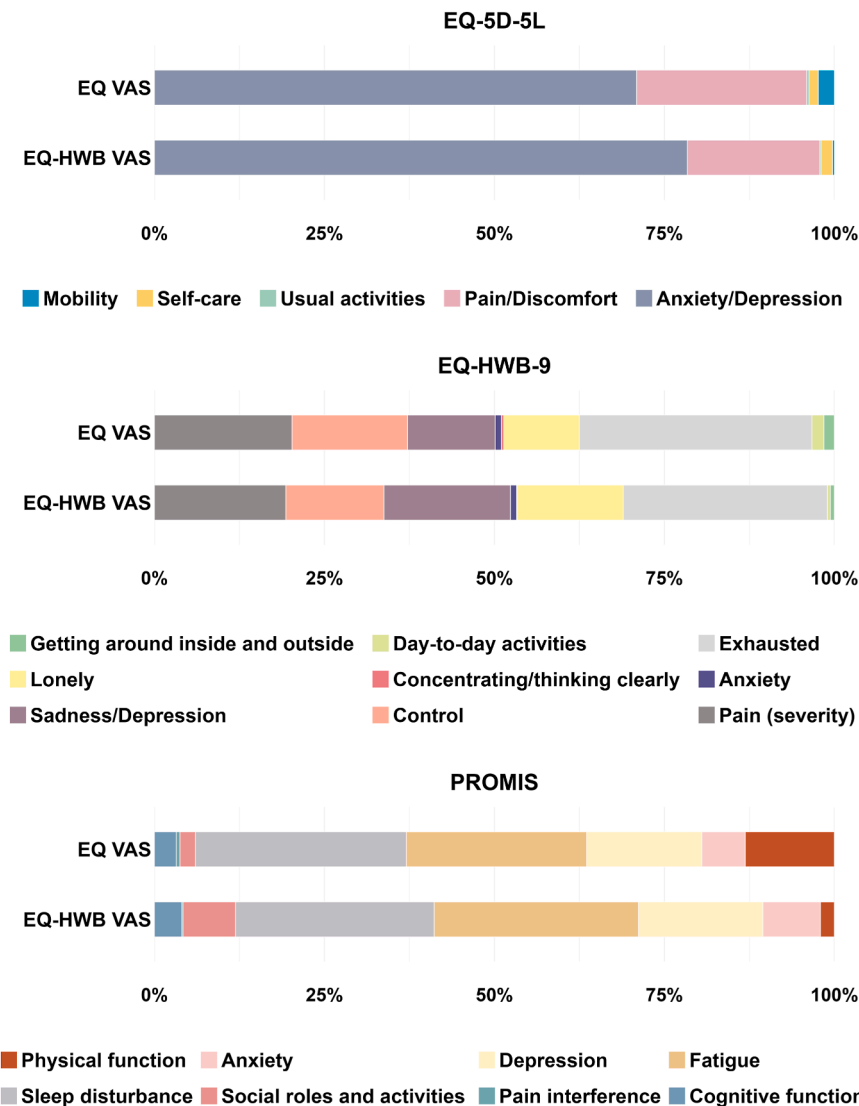
Note. Regarding the known-group analysis for the PROMIS-29+2 item "How much did pain interfere with your day-to-day activities?" Because of the small number of "very much" responses ($n = 7$), this level was combined with "quite a bit" for analysis.

Introducing the term "well-being" to the label and endpoints further extends the scope of the VAS, possibly introducing additional variability in responses due to the even larger subjectivity and the lack of a widely accepted definition of well-being.³⁷⁻³⁹ The "health and well-being" label may be more prone to reflect external factors (eg, social, economic, environmental aspects) rather than individual ones.⁴⁰ Another potential issue, in addition to individual interpretations, is that it creates a double-barreled question ("health and well-being"). It is unclear whether respondents consider both terms or only 1, and if so, which one when completing the EQ-HWB VAS.

Overall, our findings show only minor differences in measurement performance between 2 VASs, with very strong convergent validity and good agreement with each other. Some of the results suggest slightly better sensitivity of EQ-HWB VAS for non-health aspects, but the difference is very small. The EQ-HWB VAS showed a minimally lower ceiling and was less correlated with the EQ-5D-5L LSS or utilities, potentially indicating that it measures something beyond health. It had a very similar known-group validity but slightly better performance for variables such

as caregiver status and perceived financial situation. However, the EQ VAS was somewhat better at differentiating between most chronic disease groups. Notably, the EQ-HWB VAS performed very well at differentiating between health and those with anxiety or depression, but again, the ES difference was small.

The first limitation of this study is the use of a relatively healthy general population sample from one country. For instance, very few participants reported problems on certain EQ-5D-5L dimensions (eg, mobility and self-care) or EQ-HWB-9 items (eg, getting around inside and outside). This may not fully reflect experiences of patient groups, caregivers, or social care users, where the EQ-HWB VAS's sensitivity to diverse individual experiences might differ significantly. This likely explains why these items did not significantly contribute to the observed VAS scores in the Sherrock-Shapley decomposition analysis. Second, VAS responses may be influenced by preceding items and potential ordering effects. It is unknown if results would differ after the full EQ-HWB form, and despite randomization, the second, third, and fourth VAS might have been affected by the first. Although VAS presentation order was randomized, it was not recorded in the database,

Figure 2. Results of Shorrocks-Shapley decomposition analysis.

precluding analysis of potential order effects. Third, the EQ-HWB-9 and PROPr utilities were computed using nonlocal value sets, which may not fully capture cultural or regional preferences for health and well-being, potentially introducing bias. Finally, the Traditional Chinese translation of “health and well-being” as “physical and mental health” (because of no direct equivalent for “well-being”) was confirmed via cognitive debriefing, but may limit the generalizability of our findings, highlighting the need for further testing the UK English source version and other language versions in which “well-being” can be directly translated.

Conclusions

This study provides the first empirical evidence of the measurement properties of a “health and well-being” VAS for EQ-HWB, demonstrating its good validity. These findings from Hong Kong support the value of potential addition of a VAS to the EQ-HWB. Further qualitative and quantitative work is recommended, in different languages, countries, and populations to inform decisions about the addition of a VAS to the EQ-HWB-9 and EQ-HWB.

Author Disclosures

Author disclosure forms can be accessed below in the [Supplemental Material](#) section.

Supplemental Material

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